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Controlling the Parameterized Multi-channel Wiener Filter using a tiny neural network

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***Abstract*—Noise suppression and speech distortion are two important aspects to be balanced when designing multi-channel Speech Enhancement (SE) algorithms. Although neural network models have achieved state-of-the-art noise suppression, their non-linear operations often introduce high speech distortion. Conversely, classical signal processing algorithms such as the Parameterized Multi-channel Wiener Filter (PMWF) beamformer offer explicit mechanisms for controlling the suppression/distortion trade-off. In this work, we present NeuralPMWF, a system where the PMWF is entirely controlled using a low-latency, low-compute neural network, resulting in a low-complexity system offering high noise reduction and low speech distortion. Experimental results show that our proposed approach results in significantly better perceptual and objective speech enhancement in comparison to several competitive baselines using similar computational resources.**

1. INTRODUCTION

Multi-channel Speech Enhancement (MCSE) and beamforming are fundamental tasks in audio signal processing. They consists of estimating a target speaker's sound from a set of noisy microphone array measurements. Example applications of MCSE are voice capture systems, such as voice assistants, tele-conference applications, hearing aids and wearables, such as smart glasses [1], [2]. Effective SE methods must remove noise while preserving speech inteligibility and quality, that is, without introducing speech distortion [3]. This task is particularly challenging under environments containing low Signal-to-Noise Ratio (SNR), moving sources, babble noise, reverberation, among other effects.

SE methods may be categorized as Digital Signal Processing (DSP)-based, Deep Neural Network (DNN)-based, or hybrid. Many DSP-based methods are typically based on linear filters, and received significant attention for more than five decades [4]–[7]. An important aspect when choosing a beamformer is its noise suppression vs speech distortion trade-off. A popular high suppression/distortion beamformer is the Multi-channel Wiener Filter (MWF) [8], [9], while popular low distortion ones are the Minimum Variance Distortionless Response (MVDR) beamformer [10]–[12] and the Generalized Sidelobe Canceller (GSC) [13]. This work gives particular focus to the Parameterized Multi-channel Wiener Filter (PMWF) [3], [14], a generalization of the three aforementioned beamformers which allows for the beamformer's suppression/distortion pattern to be controlled for each frequency band. Although dynamically controlling this pattern offers great potential in making the filter adapt to different scenarios, the only works we found to do so were [15]–[17], which rely on specialized parameter tuning, which hinders the method's applicability in practical scenarios.

Conversely, many DNN-based methods have also been proposed for multi-channel SE [18]–[21], where multiple non-linear operations result in excellent noise suppression at the cost of speech distortion. Furthermore, neural methods often incur high memory and CPU usage, making them unsuitable for portable devices such as smart glasses. This paper focuses on hybrid methods [9], [22]–[25], which jointly utilize DSP and DNN modules, allowing for systems with better noise suppression than pure DSP and lower speech distortion and complexity than end-to-end DNN systems.

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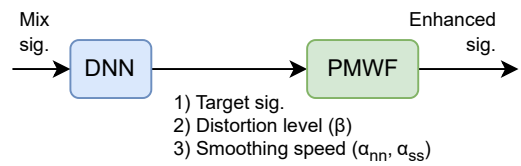

Fig. 1: Overview of the proposed NeuralPMWF MCSE method.

As shown in Fig. 1, we propose NeuralPMWF, a hybrid MCSE model employing a small DNN to fully control the PMWF. The PMWF requires estimation of speech and noise covariance matrices, which have been classically estimated using DSP methods, usually involving Speech Presence Probability (SPP) [26]. In contrast, we estimate these statistics from a multi-channel target signal estimate produced by a DNN, from which covariance matrices are estimated using exponential smoothing. Our system also estimates the required parameters α_{ss} and α_{nn} for smoothing, which control the speed to which the filter adapts to acoustic changes. Finally, our filter dynamically controls the PMWF's distortion parameter β, which we show to result in a significant performance improvement.

The main contribution of this paper is therefore a novel low-latency and low-compute multi-channel speech enhancement system using a small neural network and the PMWF, along with two ablation studies, and comparisons with state-of-the-art baselines. This paper continues as follows. In Sec. 2, the signal model and system goal are described. In Sec. 3, the proposed NeuralPWMF method is described. In Sec. 4, we perform an extensive ablation study of the proposed approach, followed by a comparison with competitive baselines. Finally, we analyze the results in Sec. 5 and conclude the work in Sec. 6.

2. SIGNAL MODEL, NOTATION AND GOAL

We represent the multi-channel speech signal as $\mathbf{S}$, the received signal by the microphone array as $\mathbf{Y}$, and the additive multi-channel noise, $\mathbf{N}$, all represented as tensors in Short Time Fourier Transform (STFT) domain as

$$\mathbf{Y} = \mathbf{S} + \mathbf{N} \in \mathbb{C}^{T \times F \times M}, \tag{1}$$

where T, F are the total number of time and frequency bins of interest, and M is the number of available microphone channels. Note that although all time bins are expressed in (1), the systems analyzed in this paper are causal, processing each time bin using past

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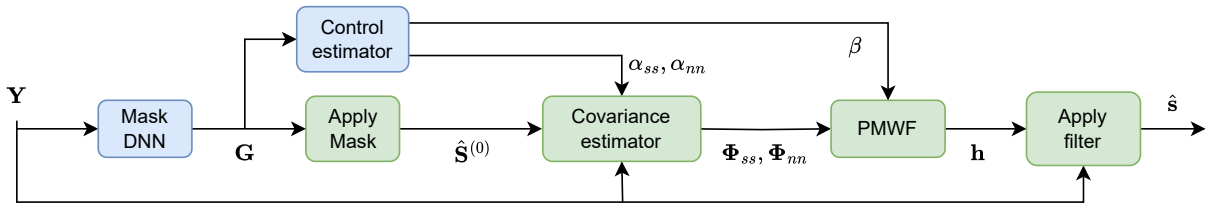

Fig. 2: System overview. DNN blocks are coloured in blue, and DSP blocks are coloured in green.

information only. We use a programming-inspired tensor indexing to access specific time/frequency/channel bins in (1). Namely, we define the goal of a multi-channel SE system as estimating the source signal at reference channel 0, that is, $\mathbf{S}_0 = \mathbf{S}[:,:,0]$, where : refers to all elements within the axis of interest. We shall also use the shorthand $\mathbf{X}[t,w] = \mathbf{X}[t,w,:]$ when accessing all microphone channels of particular Time-Frequency (TF) bins of $\mathbf{X}$.

3. PROPOSED METHOD: NEURAL PMWF

The system diagram is shown in Fig. 2. The goal of our system is to estimate a TF varying filter $\mathbf{h}[t,w] \in \mathbb{C}^M$ that defines a combination of all channels signals, that is

$$\hat{s}[t,w] = \mathbf{h}[t,w]^T \mathbf{S}[t,w]. \tag{2}$$

The motivation of employing linear filtering in favour of end-to-end neural systems is introducing meaningful signal processing-based inductive biases, which result in lower speech distortion levels on smaller networks. In our application, the filter $\mathbf{h}[t,w]$ (2) is the PMWF, which was chosen for its explicit ability of controlling speech distortion. It is derived in [3] using Lagrange multipliers as

$$\begin{aligned} \boldsymbol{\gamma}[t,w] &= \mathbf{\Phi}_{nn}^{-1}[t,w]\mathbf{\Phi}_{ss}[t,w] \in \mathbb{C}^{M\times M} \\ \mathbf{h}[t,w] &= \frac{\boldsymbol{\gamma}[t,w][:,0]}{\beta[t,w] + \text{trace}(\boldsymbol{\gamma}[t,w])}, \end{aligned} \tag{3}$$

Where variable $\boldsymbol{\gamma}$ is defined to improve the formula's legibility. We first note that the formula requires the speech and noise covariance matrices, defined as $\mathbf{\Phi}_{xx} = \mathbb{E}(\mathbf{X}[t,w]\mathbf{X}[t,w]^H) \in \mathbb{C}^{M\times M}$, where $\mathbf{X} = \{\mathbf{S}, \mathbf{N}\}$. Latent signals $\mathbf{S}$ and $\mathbf{N}$ are estimated from mixture using a DNN described in Sec. 3.2. The numerator $\boldsymbol{\gamma}[t,w][:,0]$ corresponds to choosing reference channel 0 of $\boldsymbol{\gamma}$. The denominator is a sum of the trace, i.e., the sum of $\boldsymbol{\gamma}$'s diagonal elements with the parameter β, which controls the filter's output speech distortion and is central to this work.

An essential step for computing the filter coefficients using (3) consists of estimating the speech and noise covariance matrices. In this work, we use a network to learn a multi-channel complex-valued TF mask, which is then used to estimate a multi-channel target speech signal $\hat{\mathbf{S}}_0 \in \mathbb{C}^{T\times F\times M}$, which is then used to estimate noise as $\hat{\mathbf{N}}_0 = \mathbf{Y} - \hat{\mathbf{S}}_0$ (using (1)). Finally, $\hat{\mathbf{S}}_0$ and $\hat{\mathbf{N}}_0$ are used to obtain the final covariance matrices through exponential smoothing as

$$\begin{aligned} \mathbf{\Phi}_{ss}[t,w] &= (1-\alpha_{ss}[w])\mathbf{\Phi}_{ss}[t-1,w] \\ &\quad + \alpha_{ss}[w](\hat{\mathbf{S}}_0[t,w]\hat{\mathbf{S}}_0[t,w]^H) \\ \mathbf{\Phi}_{nn}[t,w] &= (1-\alpha_{nn}[w])\mathbf{\Phi}_{ss}[t-1,w] \\ &\quad + \alpha_{nn}[w](\hat{\mathbf{N}}_0[t,w]\hat{\mathbf{N}}_0[t,w]^H), \end{aligned} \tag{4}$$

where $0 < \alpha_{ss}, \alpha_{nn} < 1$ control the filter's adaptation speed. In our work, we learn optimal frequency-dependent αs during training, as we found it advantageous to allow some bins to change faster than others. Works such as [26]–[28] propose to make the α's TF varying, which we include as model variants in our ablation studies. Finally, many works set the distortion control parameter β as 0 or 1 as to obtain the classical MVDR and MWF filters respectively. In contrast, we will show that dynamically controlling β results in expressive performance gains. The proposed strategies for controlling α and β are described below.

3.1. Control estimation

Our proposed method operates in two steps. First, a DNN estimates a multi-channel target mask $\mathbf{G}$, which is used both for estimating target signal $\hat{\mathbf{S}}_0$ and filter distortion controls β. Furthermore, α_{ss} and α_{nn} are jointly learned during training, not requiring mask $\mathbf{G}$ for their estimation. We define the controls α_{nn}, α_{ss} and β as

$$\begin{aligned} \hat{p}[t,w] &= \text{sigmoid}(\mathbf{p}^{(a)}[w]|\mathbf{G}[t,w,0]| + \mathbf{p}^{(b)}[w]) \\ \beta[t,w] &= \beta^{(0)}[w](1-\hat{\mathbf{P}}[t,w]) \\ \alpha_{ss}[w] &= \text{sigmoid}(\alpha_{ss}^{(0)}[w]) \\ \alpha_{nn}[w] &= \text{sigmoid}(\alpha_{nn}^{(0)}[w]). \end{aligned} \tag{5}$$

$\hat{p}$ can be interpreted as the Speech Presence Probability (SPP) [26], which is frequently used as an intermediary feature in SE. It is computed using the mask applied to reference channel 0. $\mathbf{p}^{(a)}, \mathbf{p}^{(b)}, \alpha_{nn}^{(0)}, \alpha_{ss}^{(0)}, \beta_{ss}^{(0)} \in \mathbb{R}^F$ are modeled to act as time-invariant gains to each control parameter, and are learned during training along with the mask network. Note that the operations in (5) are not to be confused with fully connected layers, which were found to be more computationally expensive and unstable during training.

3.2. Network architecture

The network is divided in two blocks, the spatial and temporal processing blocks, which have the goals of extracting patterns in channel and time dimensions respectively.

As a first step, the real and imaginary parts of the complex-valued spectrogram are decoupled into separate channels, resulting in a real-valued tensor of dimensions $2M \times T \times F$. This tensor is then processed using a sequence of convolutional layers called spatial processing blocks, introduced in [29], [30], which mimic a frequency domain Filter-And-Sum [6] beamformer operation. Each spatial convolution layer l comprises F distinct matrices, which are multiplied by the input tensor at each frequency bin. Each convolution is followed by parametric ReLU non-linearity. All but the last set of spatial convolutions have the same input/output size except for the last one, which outputs an additional channel which is used as the temporal block inputs. This is expressed in Fig. 3 as two separate blocks to improve comprehension.

The temporal block consists of a sequence of SplitGRU units introduced in [31], which provide an efficient alternative to classical Gated Recurrent Unit (GRU) [32] units. The additional channel generated from the last spatial processing layer is encoded to a

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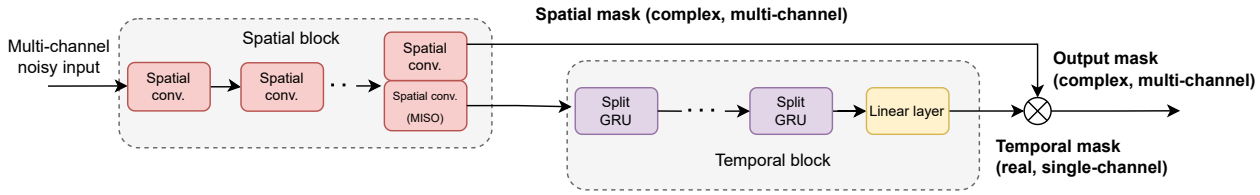

Fig. 3: Proposed network architecture for the MaskDNN block in Fig. 2, which estimates the complex-valued mask $\mathbf{G}$.

feature space of size H by applying a linear layer. These features are directed into a series of causal (uni-directional) SplitGRU layers, each with a hidden size of H. The SplitGRU layer differs from conventional GRU layers by dividing the input into R segments across the feature dimension and processing each segment with one of R parallel GRUs. The outputs of each layer are reorganized such that the output from a specific GRU is distributed to all GRUs in the subsequent layer. Employing a split factor of R effectively reduces computational demands by a factor of R. The final output of the GRU model is then projected to a size of F using a linear layer.

Finally, the multi- and single-channel outputs from the spatial and temporal blocks are combined into a multi-channel, complex-valued mask $\mathbf{G}$ by treating the temporal block's output as a real-mask itself, which is independently applied to each channel of the spatial block's output.

4. EXPERIMENTATION

We perform three experiments, two of which are ablation studies and the remaining one is a comparison of our best performing model and with baselines. The multichannel time-domain signal is normalized to a range between -60 dB and -20 dB at the reference microphone. All models are trained using Pytorch [33] with a time-domain SNR loss and a frequency-domain phase constrained magnitude (PCM) loss proposed in [34]. Training spans 100 epochs with 10- second utterances and a batch size of 128, distributed across multiple Nvidia H100 GPUs. We employ the Adam optimizer [35], with AMSGrad [36], and clipped gradient norms at 1. The learning rate starts at 0.001 for the first 70 epochs and is reduced by a factor of 10% every 10 epochs thereafter. Model evaluations utilized were Short-Time Objective Intelligibility (STOI) [37], narrow-band Perceptual Evaluation of Speech Quality (PESQ) [38], and SNR w.r.t. the reverberant target speech [39].

The MaskDNN architecture used consists of four spatial convolution filters, where the first 3 maintain the input channel size at 10, representing the real and imaginary parts of the spatial features. The temporal block contains 3 SplitGRU layers with 96 hidden units and 2 splits each. STFT settings include 256-sample window size, 128-sample frame-shift, and 129 frequency bins, therefore characterizing the NeuralPMWF as having an algorithmic latency of 16 ms to process one audio frame at 16kHz [9].

4.1. Dataset

To create pairs of clean and noisy signals for training, we utilize the Interspeech 2020 DNS Challenge corpus [40]. The speakers and noises in the training set are randomly allocated into training, testing, and validation subsets with split ratios of 85%, 5%, and 10%, respectively. All utterances are resampled to 16 kHz prior to data generation. Initially, random room dimensions are determined: length and width are uniformly sampled from [3, 10] meters, and height from [2, 5] meters. We then randomly select the location and orientation of the microphone within the room. The speech source is positioned at a distance ranging from [0.5, 2.5] meters from the array center within a Field-of-View (FoV) of $[-30°, 30°]$ azimuth and $[-90°, 90°]$ elevation, with the goal of evaluating performance on smart glasses applications where the source of interest lies in front of the wearer. Between 1 and 10 noise sources are placed at distances exceeding 0.5 meters from the array center. Between 0 and 10 interfering talker locations are positioned at distances greater than 3 meters. The Signal-to-Noise Ratio (SNR) is sampled from [-5, 10] dB, and the Signal-to-Interference Ratio (SIR) from [5, 10] dB. Data is simulated using the image method [41] using Pyroomacoustics [42] with order 6 and a uniformly sampled wall absorption coefficient from [0.1, 0.7]. To simulate multichannel data, we auralize the simulated sources to a 5-microphone array resembling Rayban Meta smart glasses by utilizing measured anechoic Acoustic Transfer Functions (ATFs), simulating the effects of channel directivity, head diffraction and absorption. In total, we simulate 320k, 600 and 3.2k utterances for training, validation, and testing respectively.

4.2. Ablation studies: controlling smoothing and distortion

To obtain the model described in Sec. 3, we performed two ablation studies focused on estimating the best strategies for controlling covariance smoothing (α) and distortion (β), whose results are shown in Table 2. For the α ablation study, we test four different smoothing techniques: (i) cumulative mean [12], (ii) frequency-independent untrained exponential smoothing, (iii) frequency-dependent trained smoothing and (iv) SPP-driven exponential smoothing [27] (as described in Sec. 3.1). For the distortion control experiments we test five different conditions, with three fixed β variants: (i) $\beta = 0$ (MVDR), (ii) $\beta = 1$ (MWF) and (iii) $\beta = 10$ (Aggressive MWF), and two learned variants: (iv) frequency-dependent and (v) the SPP-driven β proposed in Sec. 3.1.

Note that the diagram in Fig. 2 shows that α_{ss} and α_{nn} can be estimated online, we found no advantage of doing so in our experiments, and therefore only present experiments where they are learned during training and fixed during inference.

4.3. Experiment 3: Baseline comparison

Here we compare the proposed NeuralPMWF method with several low-compute neural network baselines. Firstly, we replicated the multichannel speech-enhancement MC-CRN baseline from [43], an extension of [44] that incorporates spatial information using 20 spatially uniformly distributed maxDI beamformers [45]. MC-CRN modifies the Equivalent Rectangular Bandwidth (ERB) features of the reference channel with a real-gain. Furthermore, we adapted the low-compute GTCRN [46] to handle multichannel audio by concatenating input channels and increasing the number of input filters. Finally, include the TinyGRU+MWF baseline proposed in [29], which estimates a single-channel complex-valued mask used in conjunction with Wang et. al's MWF formulation [9].

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Method	STOI	SI-SDR	SNR	NB-PESQ	MMACs	Params.
NeuralPMWF (ours)	**74.3**	**5.5**	**6.91**	**2.12**	24.95	164.9k
TinyGRU+MWF	73.1	5.02	6.56	2.08	24.37	152.8k
GTCRN+MWF	72.7	4.66	6.23	2.01	91	26.5k
MCCRN+MWF	69.9	3.88	5.78	1.79	35	117k
Input	58.1	-2.83	-2.84	1.47	–	–

Table 1: Comparison of the proposed NeuralPMWF and different baselines. SI-SDR and SNR are expressed in dB. MMACs stands for million multiply-accumulate operations per second.

(a) α: **Smoothing Control**				
Method	**STOI**	**SI-SDR**	**SNR**	**NB-PESQ**
Cum. mean	69.8	3.62	5.57	1.93
Fixed	71.8	4.83	6.40	2.04
Frequency-dependent	**72.3**	**4.90**	**6.46**	**2.06**
SPP-driven	71.9	4.90	6.45	2.05
(b) β: **Distortion Control**				
$\beta = 0$ (MVDR)	69.8	3.62	5.57	1.93
$\beta = 1$ (MWF)	69.9	3.62	5.57	1.93
$\beta = 10$	67.2	1.71	4.42	1.87
Frequency-dependent	70.0	3.70	5.62	1.93
SPP-driven	**74.3**	**5.5**	**6.91**	**2.12**

Table 2: (a) smoothing (α_{nn} and α_{ss}) and (b) distortion (β) strategy comparison. In (a), several β strategies are tested while cumulative mean covariance smoothing was used. In (b), several α strategies are used while β was fixed at 0 (MVDR)

5. RESULTS

The baseline comparison results shown in Table 1 reveal that NeuralPMWF surpasses all baselines in all metrics, while requiring very low computational resources. Turning to the ablation study results in Table 2-a, cumulative mean showed the worst performance among all smoothing techniques, explained by its diminishing ability to adapt to acoustic changes over time. Other exponential smoothing methods performed similarly. NeuralPMWF is also attractive due to its explainability, as the learned internal parameters can be visualized, as shown in Fig. 4. In Fig. 4a, the speech smoothing parameter α_{ss} is bigger than the noise's α_{nn}, in accordance with the classical assumption that speech statistics change faster than noise [27].

Turning to the distortion control experiments shown in Table 2-b, agressive MWF ($\beta = 10$) performed the worst among all variants. The MVDR and MWF showed almost identical results. Finally, expressive performance improvements were obtained by applying SPP-driven β, demonstrating the importance of dynamic, time-varying distortion control. As shown in Fig. 4-c, very agressive filtering ($\beta > 30$) can be applied if the network is certain of speech absence. Fig. 4-b shows that learning a time-invariant frequency-dependent value for β results in low distortion for low frequencies, where most speech energy is typically located [47].

6. CONCLUSION AND FUTURE STEPS

We presented NeuralPMWF, a novel system that utilizes a tiny neural network to control the classical beamformer algorithm PMWF. Our system is trained end-to-end to estimate noise and speech spatial characteristics, while dynamically controlling the PMWF's output distortion and the covariance smoothing parameters in real-time, resulting in increased speech quality when comparing to competitive low-compute multichannel baselines. Future research directions include using alternative PMWF formulations and further optimizing the network MIMO architecture, as well as investigating the method's generalization to geometries other than the one it was trained on.

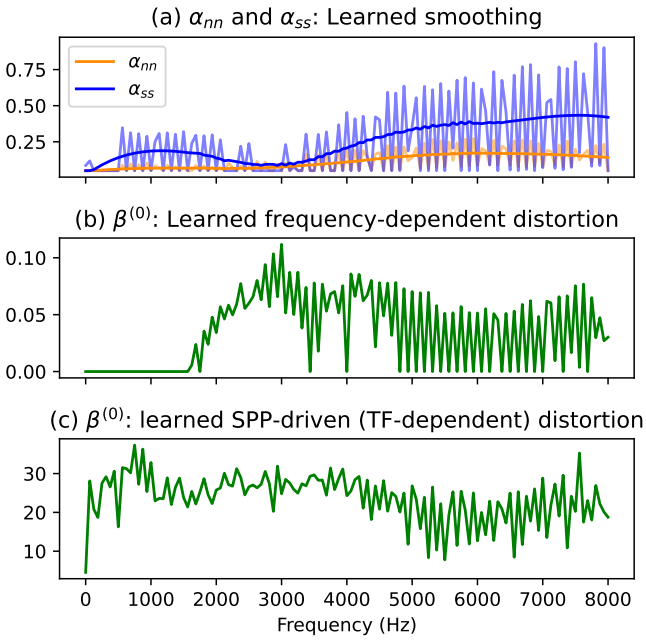

Fig. 4: (a): Learned speech and noise covariance smoothing parameters per frequency. The darker curve shows the interpolated filter using a Savitsky-Golay filter to exhibit a trend. (b): $\beta^{(0)}$ results when training using SPP-independent β. (c): $\beta^{(0)}$ for SPP-dependent $\beta^{(0)}$.

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